



IILM UNIVERSITY
Greater Noida, Uttar Pradesh

VIRTUAL INTERNSHIP REPORT

Skyscanner Software Engineering Virtual Experience Program

Presented by: Akarsh Anand

Enrollment No: 25SCS1003000760

Program: B.Tech CSE | Semester: 3rd

Organization: Skyscanner

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About the Program



Global travel search platform comparing flights, hotels & car rentals

Skyscanner

Skyscanner is a global travel search engine, headquartered in Edinburgh, Scotland, helping millions of travellers compare flights, hotels and car rentals across hundreds of providers. It is known for its strong engineering culture and its Backpack design system.

Forage

Forage partners with global companies to offer free, self-paced job-simulation tasks that mirror real day-to-day work — giving students hands-on, practical experience before entering the industry.

This internship consisted of 4 introductory tasks (1–2 hrs each), covering front-end web, backend microservices, mobile app development, and UX/UI design.

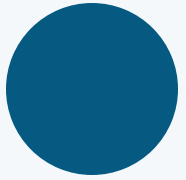
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Project Objectives

- 1 Apply Skyscanner's Backpack design system to build a functional React web application.
- 2 Design and implement a Dropwizard backend microservice for car rentals and hotels.
- 3 Develop a basic Android app that displays a user's flight itinerary using Backpack.
- 4 Create user-centric design mockups improving ease of use for developers and users.
- 5 Gain hands-on exposure to full-stack, industry-style software engineering tasks.

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Tasks Undertaken



TASK 1

Backpack React Web App

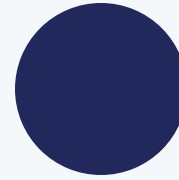
Front-end • Introductory • 1–2 hrs



TASK 2

Dropwizard Microservice

Backend • Introductory • 1–2 hrs



TASK 3

**Mobile Flight Itinerary
App**

Mobile • Introductory • 1–2 hrs



TASK 4

**User-Friendly Platform
Design**

UX/UI • Introductory • 1–2 hrs

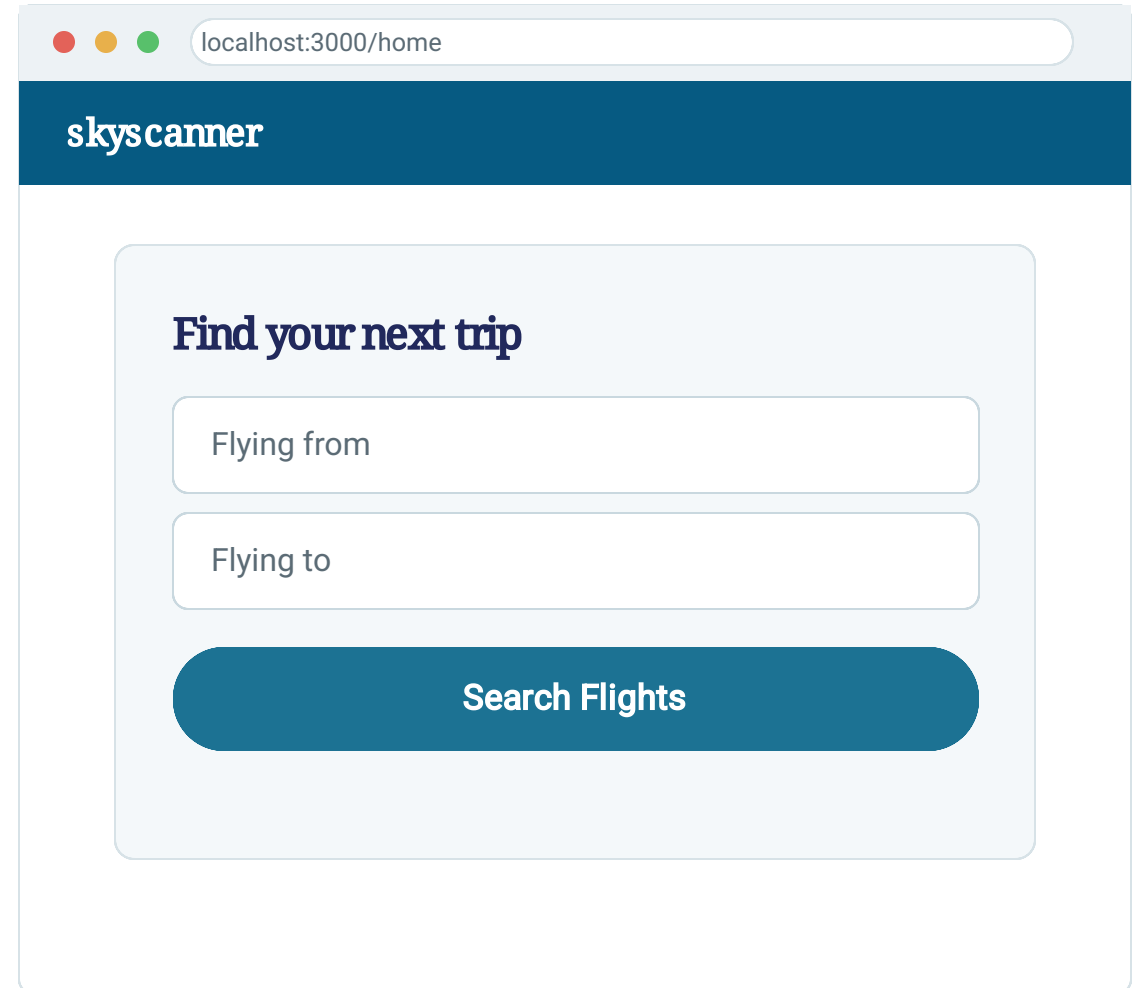
Create a Backpack React Web App

Implementation

```
import React, { useState } from 'react';
import BpkButton from 'bpk-component-button';
import BpkCard from 'bpk-component-card';
import BpkInput from 'bpk-component-input';
const HomePage = () => {
  const [origin, setOrigin] = useState("");
  const [dest, setDest] = useState("");
  const handleSearch = () => {
    console.log(`${origin} -> ${dest}`);
  };

  return (
    <BpkCard className="home-card">
      <h2>Find your next trip</h2>
      <BpkInput placeholder="From" value={origin}
        onChange={e => setOrigin(e.target.value)} />
      <BpkInput placeholder="To" value={dest}
        onChange={e => setDest(e.target.value)} />
      <BpkButton onClick={handleSearch}>
        Search Flights
      </BpkButton>
    </BpkCard>
  );
};
```

Output



localhost:3000/home

skyscanner

Find your next trip

Flying from

Flying to

Search Flights

Create a Microservice (Dropwizard)

Implementation

```
@Path("/v1/rentals")
@Produces(MediaType.APPLICATION_JSON)
public class RentalResource {

    private final RentalService rentalService;

    public RentalResource(RentalService service) {
        this.rentalService = service;
    }

    @GET
    @Timed
    public Response getRentals(
        @QueryParam("city") String city) {
        List<Rental> results =
            rentalService.findAvailable(city);

        if (results.isEmpty()) {
            return Response.status(404).build();
        }
        return Response.ok(results).build();
    }
}
```

Output

GET /v1/rentals?city=Edinburgh

Rental Search Results

200 OK

HertzCo

Edinburgh, UK

£34.50 / day

Rental ID: r-1042 | Currency: GBP

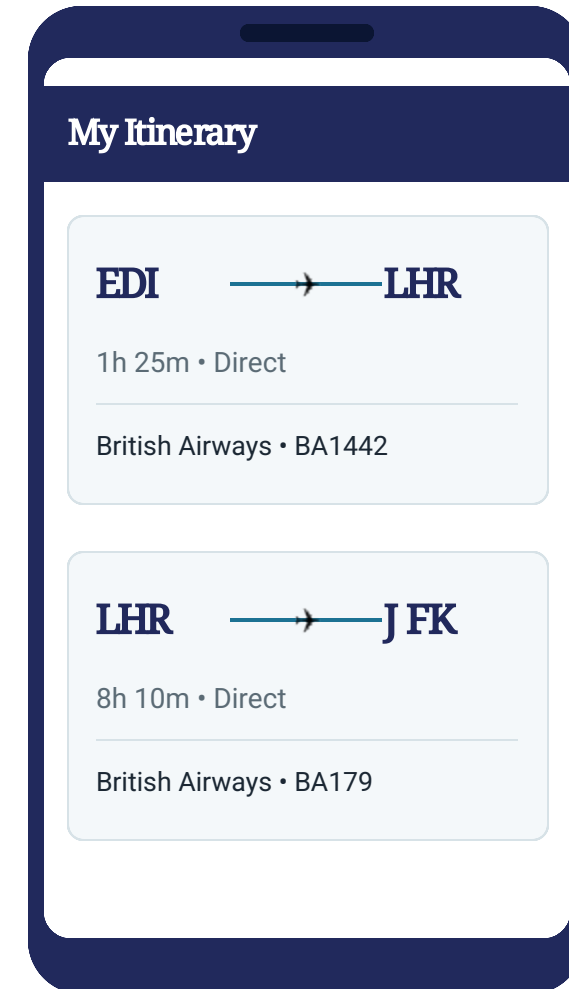
Response time: 128ms

Create a Mobile Flight Itinerary App

Implementation

```
class ItineraryAdapter(  
    private val flights: List<FlightSegment>  
) : RecyclerView.Adapter<ViewHolder>() {  
  
    class ViewHolder(view: View) :  
        RecyclerView.ViewHolder(view) {  
        val origin: BpkText =  
            view.findViewById(R.id.originCode)  
        val dest: BpkText =  
            view.findViewById(R.id.destCode)  
        }  
  
    override fun onBindViewHolder(  
        holder: ViewHolder, position: Int  
    ) {  
        val flight = flights[position]  
        holder.origin.text = flight.originCode  
        holder.dest.text = flight.destCode  
    }  
  
    override fun getItemCount() = flights.size  
}
```

Output



TASK 4

Build a User-Friendly Platform

UX / UI Design Task — creating designs that enable ease of use for the developer team.

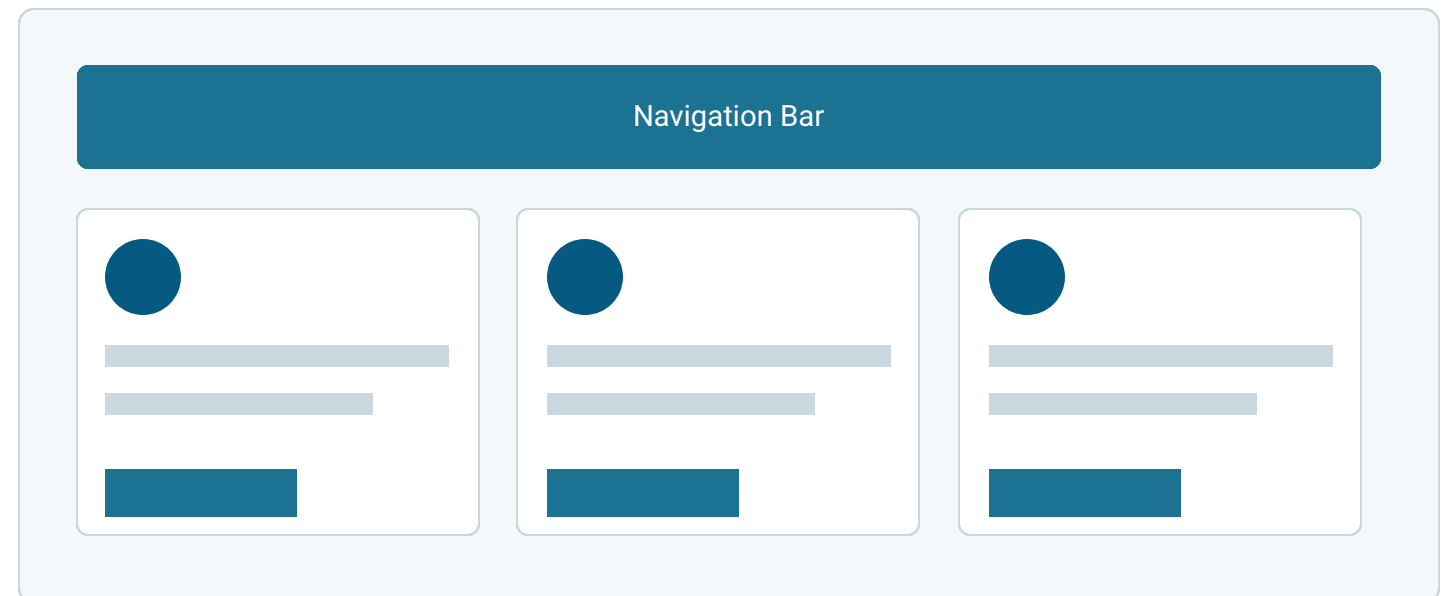
Tools Used

Figma • Backpack Design Tokens • UX Research Principles

What was done

- Analyzed usability needs from both a developer's and an end-user's perspective
- Created wireframes and mockups for a consistent, intuitive interface
- Applied design-consistency principles used across Skyscanner products
- Proposed an easy-to-implement platform layout for the dev team to adopt

Output — Wireframe



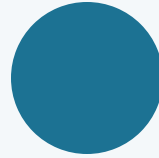
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Technologies & Tools Summary



Frontend

React.js, Backpack React,
JSX, CSS3



Backend

Java, Dropwizard,
REST APIs, Maven



Mobile

Android Studio, Kotlin,
Backpack for Android



Design & Tools

Figma, Design Tokens,
Git / GitHub

All four tasks together demonstrate a complete, cross-platform engineering workflow – from UI components to backend services to mobile delivery.

Learnings & Outcomes



Practical understanding of using an established design system (Backpack) to build consistent UIs



Hands-on experience designing and testing a REST-based backend microservice with Dropwizard



Exposure to mobile app development and displaying structured data using Android + Backpack



Improved understanding of UX/UI principles for building developer- and user-friendly platforms



Overall, a portfolio-ready demonstration of full-stack, cross-platform engineering skills

Conclusion

This Skyscanner Forage virtual internship provided practical, industry-style exposure across front-end, backend, mobile, and UX/UI design — strengthening both technical skills and understanding of how large-scale travel platforms are built.

Akarsh Anand | 25SCS1003000760 | Session 2025-29

Thank You